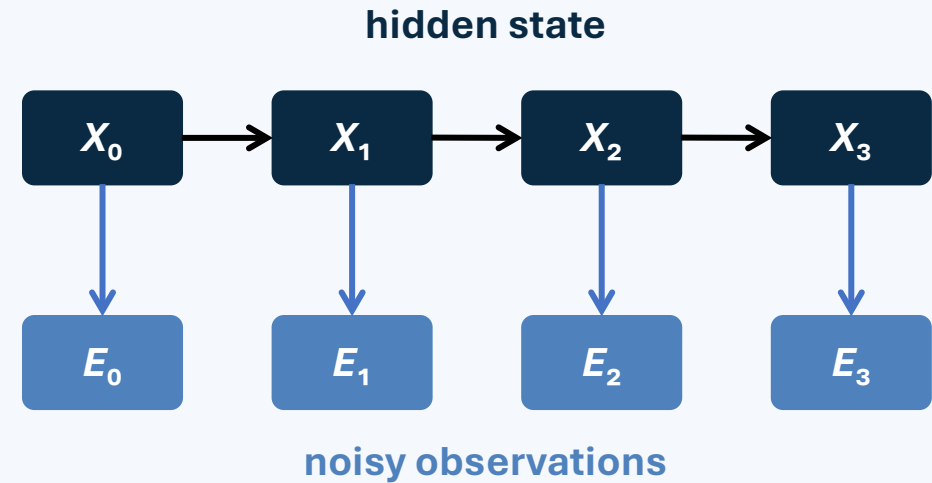


Artificial Intelligence

Lesson 23: Hidden Markov Models

Predict → Observe → Update → Repeat



An HMM tracks belief about a hidden state using transitions and noisy evidence.



The addition of an observation model

L23 adds one new ingredient to the temporal model from L22.

Markov Model

State evolves through time

$$X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow X_3$$

We propagate a belief distribution using the transition model.

Hidden Markov Model

State still evolves, but we cannot observe it directly.

$$\begin{array}{cccc} X_0 & \rightarrow & X_1 & \rightarrow & X_2 & \rightarrow & X_3 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ E_0 & & E_1 & & E_2 & & E_3 \end{array}$$

Observations provide noisy evidence about the hidden state.

The transition model stays. L23 adds an observation model.

Today's mission

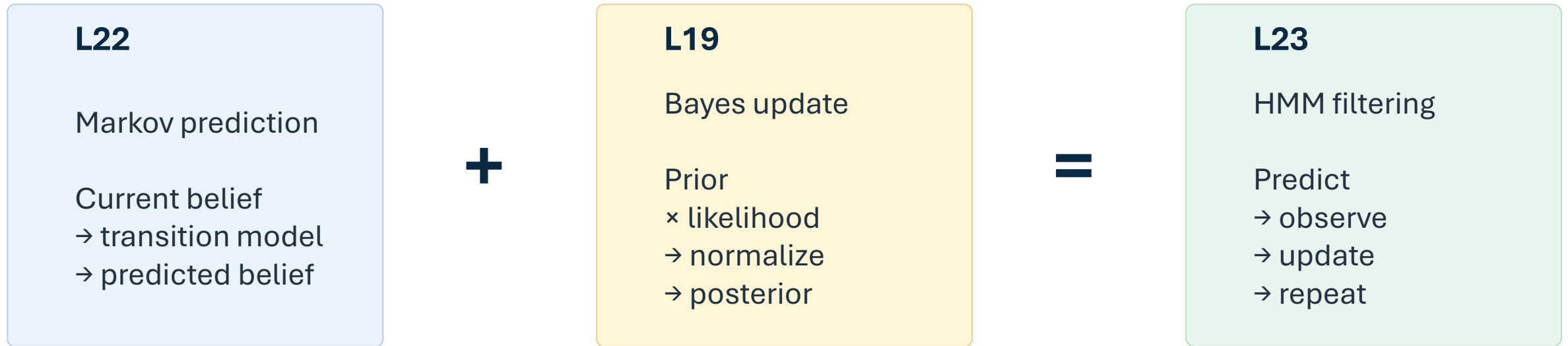
Update your belief of hidden state variables from observations.

- 1 Distinguish hidden state variables from observable evidence variables in an HMM.
- 2 Interpret the initial, transition, and observation models that define an HMM.
- 3 Perform a one-step Predict $\rightarrow$ Observe $\rightarrow$ Normalize belief update.
- 4 Interpret how sequential observations change belief about hidden states over time.

Performance target: update a hidden-state belief from a transition and a noisy sensor reading.

L22 + L19 = Hidden Markov Models

Two familiar ideas combine into one repeated inference cycle.



Today is not a new probability universe—we are combining two operations you already know.

Two kinds of variables

An HMM separates what is true from what we can observe.

Hidden state X_t

The condition we care about but cannot directly see.

Examples:

- actual system health
- actual robot location
- actual compromise state

Observation / evidence E_t

A measurable signal whose probability depends on the hidden state.

Examples:

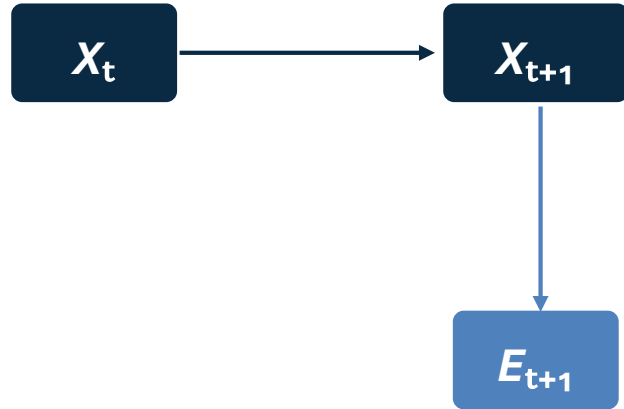
- warning light
- range-sensor reading
- IDS alert



Evidence is information about the hidden state—it is not the hidden state itself.

Persistent example: system health

We will extend the same Healthy/Degraded model from L22.



Hidden state

$$X_t \in \{ \text{Healthy}, \text{Degraded} \}$$

We maintain a probability distribution over the true condition.

Observed signal

$$E_t \in \{ \text{Alert}, \text{Normal} \}$$

The monitor is informative, but imperfect.

The only new ingredient relative to L22 is the noisy observation.

You Try: hidden or observed?

Classify each pair before we calculate anything.

Scenario	Hidden variable	Observed variable
Robot navigation		
Cyber defense		
Weather		
Equipment		

Ask: Why do we need a probability distribution over the hidden state?

What defines an HMM?

A compact HMM needs three probability models.

1. Initial distribution

$$P(X_0)$$

What did we believe about the hidden state at the start?

2. Transition model

$$P(X_{t+1} | X_t)$$

How does the hidden state evolve through time?

3. Observation model

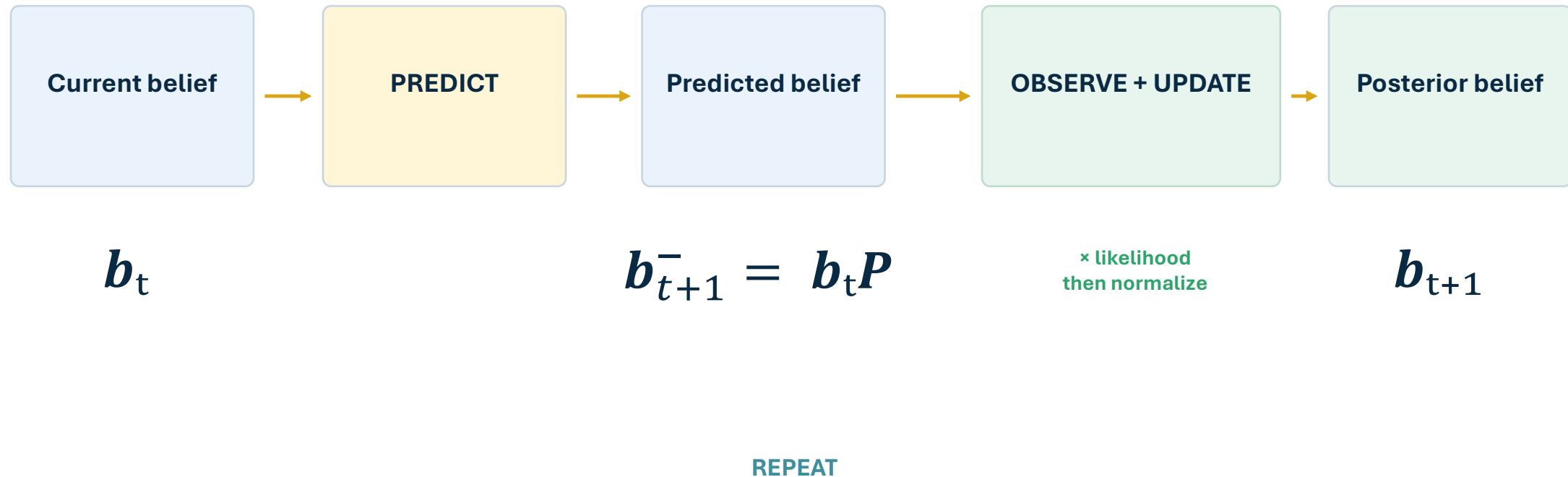
$$P(E_t | X_t)$$

How likely is each sensor reading in each hidden state?

Initial belief + state dynamics + sensor behavior = the HMM.

The HMM filtering cycle

Advance one time step by predicting first, then correcting with evidence.



Filtering asks: “What should I believe about the hidden state now, given evidence so far?”

Initial belief

Belief means a probability distribution over the hidden state.

At time t

$$b_t = [0.80, 0.20]$$

$$P(H_t) = 0.80$$

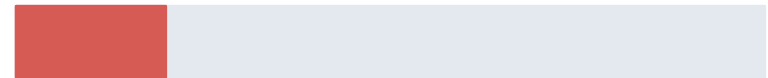
$$P(D_t) = 0.20$$

Healthy



0.8

Degraded



0.2

We do not know the state with certainty.
We maintain a belief about it.

A belief state is just a probability distribution—same idea as L18.

Transition model

This is exactly the model from L22.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Read one entry

$$P(D_{t+1} | H_t) = 0.15$$

If the system is Healthy now, it has a 15% probability of being Degraded next step.

Rows = current hidden state. Columns = next hidden state.

Observation model

The observation is noisy evidence about the hidden state.

	Alert	Normal
Healthy	0.10	0.90
Degraded	0.70	0.30

$$P(\text{Alert} \mid \text{Degraded}) = 0.70$$

An alert is much more likely if the system is Degraded—but alerts can still occur when Healthy.

$$P(\text{Alert} \mid \text{Degraded}) \neq P(\text{Degraded} \mid \text{Alert})$$

L19 returns: likelihood is not the posterior.

You Try: read the observation model

Interpret the sensor table before we update the belief.

	Alert	Normal
Healthy	0.10	0.90
Degraded	0.70	0.30

- What is $P(\text{Alert} \mid \text{Healthy})$?
- What is $P(\text{Normal} \mid \text{Degraded})$?
- Which observation provides stronger evidence for Degraded?
- Does an Alert guarantee that the system is Degraded?

Focus on interpretation, not Bayes arithmetic yet.

Step 1: Predict

Start from the current belief before the next sensor reading arrives.

Current belief

$$b_t = [0.80, 0.20]$$

Healthy 0.80
Degraded 0.20



Question

What should we believe at time t+1
BEFORE seeing the next observation?

Prediction uses only the transition model.

Predict the next belief

Everything on this slide is just L22.

$$b_t = [0.80, 0.20]$$

Healthy: $(0.80)(0.85) + (0.20)(0.30) = 0.74$
Degraded: $(0.80)(0.15) + (0.20)(0.70) = 0.26$

Predicted belief

$$b_{t+1}^- = [0.74, 0.26]$$

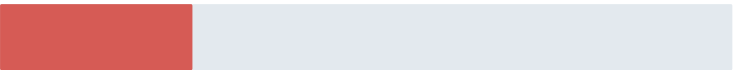
	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Healthy



0.74

Degraded



0.26

	Alert	Normal
Healthy	0.10	0.90
Degraded	0.70	0.30

The superscript “-” means predicted belief before the new observation.

Interpret the prediction

Time changes our belief even before new evidence arrives.



Why did Degraded increase?

The transition model allows a Healthy system to degrade and a Degraded system to persist. The predicted belief accounts for both pathways.

Prediction uses only the transition model.

Step 2: Observe

Now a sensor reading arrives.

Predicted belief

Healthy 0.74
Degraded 0.26

+

New evidence

$E_{t+1} = \text{Alert}$

$P(\text{Alert} \mid H) = 0.10$

$P(\text{Alert} \mid D) = 0.70$

**Before calculating:
should belief in Degraded increase or decrease?**

Predict the direction first. Then calculate the size of the update.

Bayes return

Use the predicted belief as the prior for the sensor update.

L19 pattern

Posterior $\propto$ Likelihood $\times$ Prior

Then normalize.

HMM observation update

Updated belief $\propto$ Observation likelihood $\times$ Predicted belief

Then normalize.

The “prior” in this Bayes update is the belief predicted from the transition model.

Reweight by the observation likelihood

How compatible is the Alert with each possible hidden state?

State	Predicted belief	P(Alert state)	Weight
Healthy	0.74	0.10	0.074
Degraded	0.26	0.70	0.182

$$0.74 \times 0.10 = 0.074$$

$$0.26 \times 0.70 = 0.182$$

These are evidence-weighted values, not yet a probability distribution.

Normalize the weights

Turn the evidence-weighted values back into a probability distribution.

$$\text{Total weight} = 0.074 + 0.182 = 0.256$$

Healthy posterior

$$0.074 / 0.256 \approx 0.289$$

Degraded posterior

$$0.182 / 0.256 \approx 0.711$$

$$\text{Updated belief: } b_{t+1} \approx [0.289, 0.711]$$

Is this a probability distribution?

What each operation is doing

Use the question behind each step to debug your reasoning.

PREDICT

What might the hidden state have become?

Use the transition model.

Sum over the possible previous states.

OBSERVE / REWEIGHT

How compatible is the new reading with each possible state?

Use the observation likelihood.

Multiply.

NORMALIZE

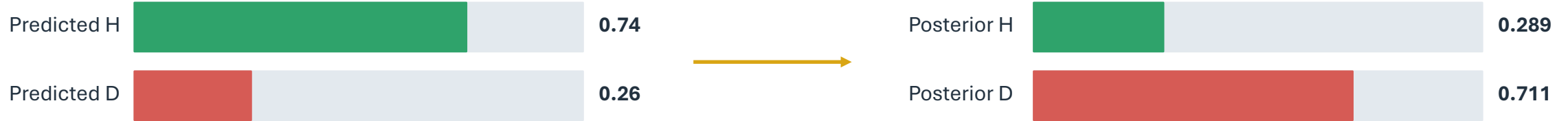
Turn the evidence-weighted values into a probability distribution.

Divide by the total weight.

Predict = sum pathways. Observe = multiply likelihood. Normalize = restore a distribution.

Compute → Interpret → Compare

The number matters only if we can explain what changed and why.



Compute

$$P(D_{t+1} | \text{Alert}) \approx 0.711$$

Interpret

After the alert, the system is believed to be Degraded with about 71% probability.

Compare

Belief in Degraded rose from 0.26 to 0.711 because Alert strongly favors D.

An observation changes belief according to how diagnostic that observation is.

You Try: complete one filtering step

Work in pairs. Show all three stages.

Current belief

$$b_t = [0.60, 0.40]$$

	H next	D next
H	0.80	0.20
D	0.25	0.75

Observation model

$$P(\text{Warning} \mid H) = 0.20$$

$$P(\text{Warning} \mid D) = 0.80$$

Observed: Warning

Your tasks

1. Predict the next belief.
2. Reweight by Warning.
3. Normalize.
4. Interpret the final $P(D)$.

You Try: complete one filtering step (solution)

Use the same Predict → Observe → Normalize structure every time.

Current belief

$$b_t = [0.60, 0.40]$$

	H next	D next
H	0.80	0.20
D	0.25	0.75

Observation model

$$P(\text{Warning} \mid H) = 0.20$$

$$P(\text{Warning} \mid D) = 0.80$$

Observed: Warning

Your tasks

1. Predict the next belief.
2. Reweight by Warning.
3. Normalize.
4. Interpret the final $P(D)$.

1. Predict

$$H: (.60)(.80) + (.40)(.25) = .58$$

$$D: (.60)(.20) + (.40)(.75) = .42$$

Predicted: [.58, .42]

2. Reweight

$$H: (.58)(.20) = .116$$

$$D: (.42)(.80) = .336$$

$$\text{Total} = .452$$

3. Normalize + interpret

$$H: .116 / .452 \approx .257$$

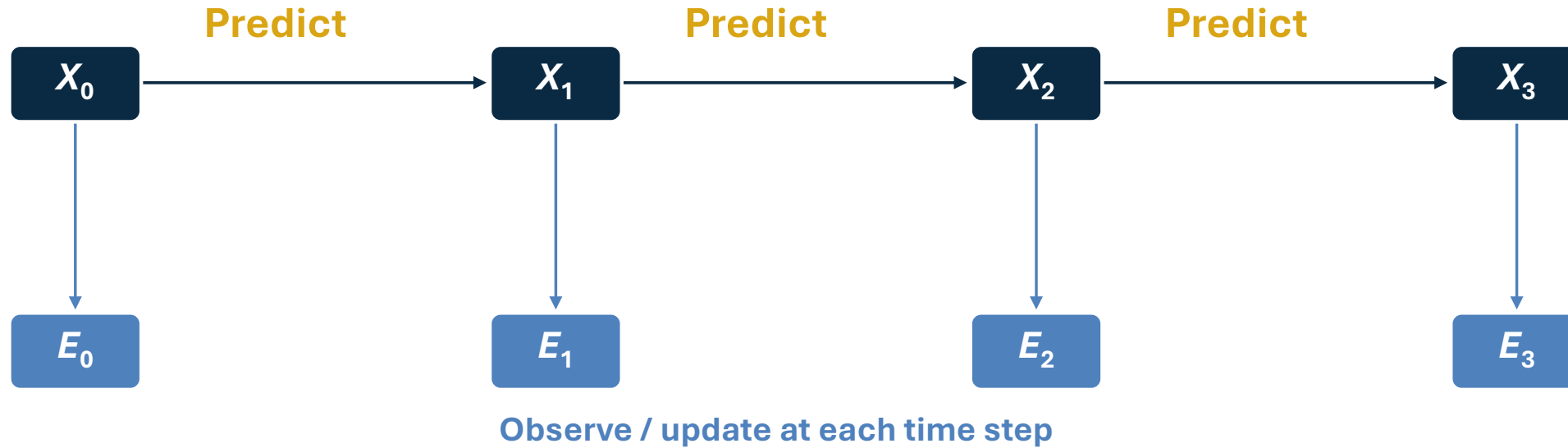
$$D: .336 / .452 \approx .743$$

A Warning shifts belief strongly toward Degraded.

The observation did not create the prior belief; it corrected the prediction produced by the transition model.

Evidence accumulates through time

Filtering repeats as the hidden process evolves and new readings arrive.



What changes over time?

Passage of time can move/spread belief; observations can sharpen or redirect it.

A posterior becomes the starting belief for the next prediction.

Sequence matters

Different evidence histories can produce different current beliefs.

Sequence A

Alert → Alert → Normal

Sequence B

Normal → Normal → Alert

Which sequence should generally leave greater belief in Degraded?

Do not calculate. Predict and justify using recency + accumulated evidence.

Evidence is interpreted in the context of the transition dynamics and all prior evidence.

Three different HMM questions

Same model, different inference goals.

FILTERING

What is the hidden state now, given evidence so far?

$$P(X_t | e_{1:t})$$

This lesson: MASTER

PREDICTION

Given current evidence, what might the hidden state be later?

Propagate the current posterior without a new observation.

MASTER conceptually

MOST-LIKELY SEQUENCE

What hidden-state path most likely explains the observations?

Different problem from filtering.

Conceptual exposure only

Be precise about the question before choosing an inference method.

Advanced HMM methods (conceptual only)

Know what problem each method addresses; execution is not required.

Viterbi algorithm

Goal: most likely hidden-state sequence.

Instead of summing over previous states, Viterbi keeps the best predecessor for each state and later reconstructs the best path.

You do not need to execute it in this course.

Particle filtering

Goal: approximate belief when the state space is too large for exact inference.

Represent the belief with many samples (“particles”), move them through the transition model, then reweight/resample using observations.

Conceptual exposure only.

Filtering is exact for our small classroom HMMs; advanced methods become useful as models grow.

Transfer: robot localization

The same HMM structure appears in many AI systems.

Hidden state

Robot location

Which cell / room / position is the robot actually in?

Transition model

Motion uncertainty

Where might the robot move next, given its current location?

Observation model

Noisy range / doorway sensor

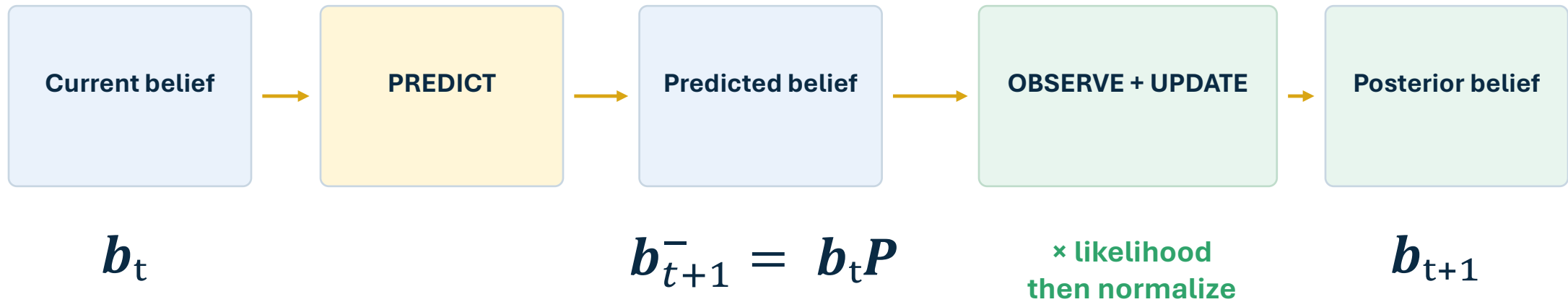
How likely is the reading from each possible location?

Quick check: during PREDICT, which model is used? During OBSERVE, which model is used?

HMMs are a natural fit when the world changes and sensors are imperfect.

Recap: the HMM filtering cycle

Advance one time step by predicting first, then correcting with evidence.



REPEAT

Markov model vs. Hidden Markov model

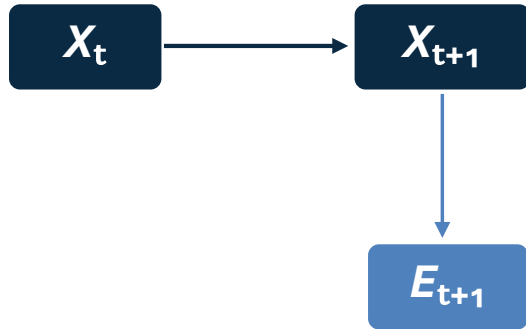
L23 adds evidence correction to L22 state propagation.

Feature	Markov model	Hidden Markov model
State evolves	Yes	Yes
Transition model	Yes	Yes
True state directly available	Assumed / represented directly	No — hidden
Observation model	No	Yes
Core operation	Predict	Predict + correct

An HMM is a Markov model plus noisy observations.

Culminating problem — predict, then update

Use the same diagram and probability tables to calculate two different values.



	S_{t+1}	F_{t+1}
S_t	0.90	0.10
F_t	0.25	0.75

	Warning	Normal
Stable	0.20	0.80
Faulty	0.80	0.20

Current belief

$$P(S_t)=0.70$$
$$P(F_t)=0.30$$

(a) Prediction

Before the next sensor reading,
calculate
 $P(F_{t+1})$.

(b) Filtering update

Then the sensor reports Warning.
Calculate
 $P(F_{t+1} \mid \text{Warning})$.

For each part, show the probability pathways you use.

Culminating problem — (a) solution

First predict the Faulty probability before any new observation.

$$P(F_{t+1}) = P(S_t)P(F_{t+1}|S_t) + P(F_t)P(F_{t+1}|F_t)$$

$$= (0.70)(0.10) + (0.30)(0.75)$$

$$= 0.07 + 0.225 = 0.295$$

Predicted distribution

Stable = 0.705 Faulty = 0.295

Part (a) is a Markov prediction: transition model only.

Culminating problem — (b) solution

Then incorporate the Warning observation.

State	Predicted belief	$P(\text{Warning} \mid \text{state})$	Weight
Stable	0.705	0.20	0.141
Faulty	0.295	0.80	0.236

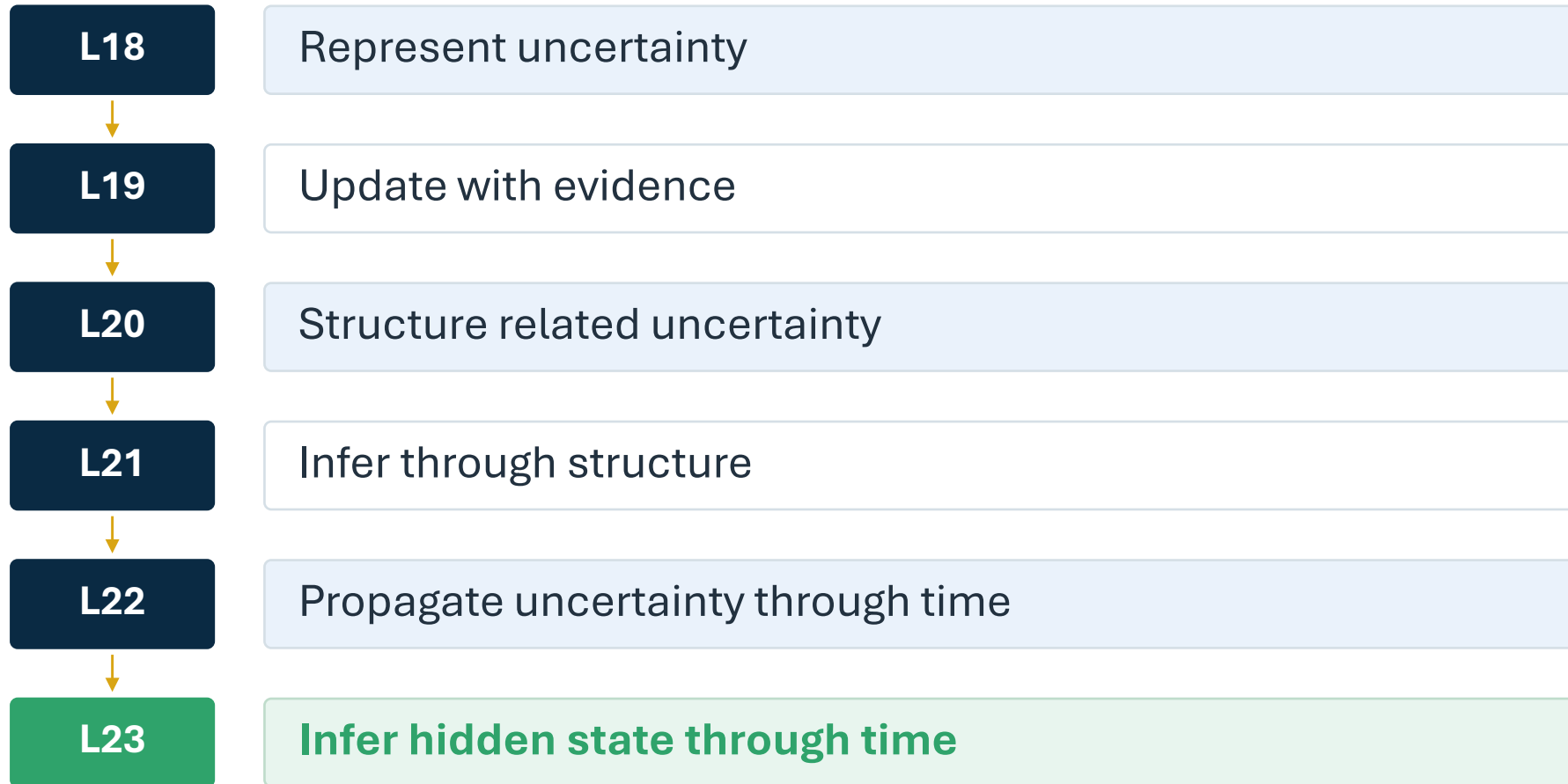
Normalize: $P(F_{t+1} \mid \text{Warning}) = 0.236 / (0.141 + 0.236) \approx 0.626$

0.295 $\rightarrow$ Warning $\rightarrow$ 0.626

Prediction asks what the state may become; filtering asks what to believe after new evidence.

The probability block comes together

L23 is the culmination of the module's reasoning tools.



The module progresses from representing uncertainty to reasoning about hidden state through time.

Three operations underlying everything

HMM filtering reuses the same probability moves from the entire block.

SUM

During prediction:

sum over possible previous hidden states.

$$\sum_x P(x'|x)b_t(x)$$

MULTIPLY

During observation:

multiply by the likelihood of the evidence.

$$P(e|x') \times \text{predicted belief}$$

NORMALIZE

After reweighting:

divide by the total so the posterior sums to 1.

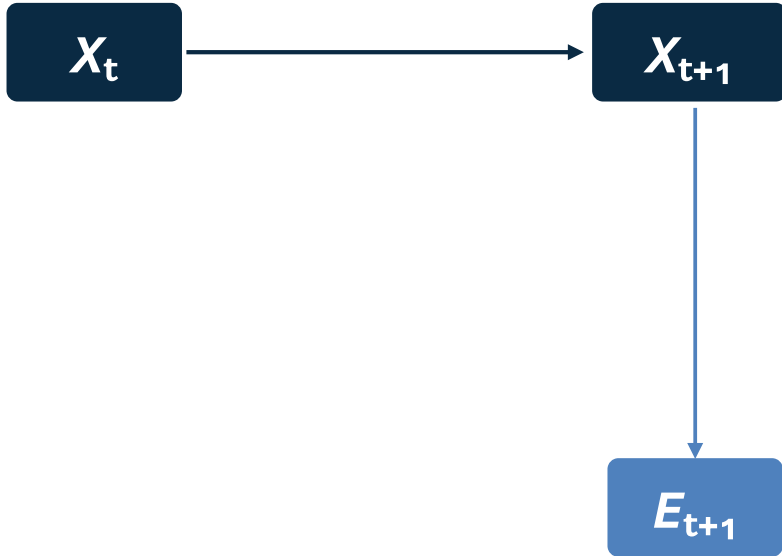
Compact forward update (literacy, not memorization):

$$\mathbf{b}_{t+1}(\mathbf{x}') \propto P(\mathbf{e}_{t+1} | \mathbf{x}') \sum_x P(\mathbf{x}' | \mathbf{x}) \mathbf{b}_t(\mathbf{x})$$

The equation packages the cycle you already understand—it is not the entry point.

Exit ticket

Can you narrate the HMM update without relying on memorized symbols?



X is the _____ variable.

E is the _____ variable.

Prediction uses the _____ model.

Observation update uses the _____ model.

After reweighting, we _____ so probabilities sum to 1.

Next: review Module 3 concepts, combining probability, Bayes, Bayes nets, Markov models, and HMMs.